

Forest fire spread simulation algorithm based on cellular automata

Xiaoping Rui¹ · Shan Hui² · Xuetao Yu³  · Guangyuan Zhang¹ · Bin Wu¹

Received: 17 July 2017 / Accepted: 14 November 2017 / Published online: 22 November 2017
© Springer Science+Business Media B.V., part of Springer Nature 2017

Abstract Traditional models result in low efficiency and poor accuracy when simulating the spread of large-scale forest fires. We constructed an improved model that couples cellular automata with an existing forest fire model to ensure better time accuracy of forest fire spread. Our model considers the impact of time steps on simulation accuracy to provide an optimal time step value. The model was tested using a case study of forest fire spread at Daxing'an Mountain in May 2006. The results show that the optimal time step for the forest fire spread geographic cellular automata simulation algorithm is 1/8 of the time taken for cellular material to be completely combusted. When compared with real fire data from Landsat Thematic Mapper (TM) images, our model was found to have high temporal and spatial consistency, with a mean Kappa coefficient of 0.6352 and mean accuracy of 87.89%. This algorithm can be used to simulate and predict forest fire spread and is also reversible (i.e., it can identify fire source points).

Keywords Forest fire spread simulation · Cellular automata · Remote sensing · Fire point identification · Forest fire disaster

1 Introduction

Forest fire spread computer simulations combine theoretical forest fire spread models with a suitable computer algorithm and display the occurrence and development of real forest fires in the form of graphics or animation. Efficient and accurate simulations can help

✉ Xuetao Yu
yuxuetaoxiaoping@163.com

¹ College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China

² Land Surveying, Planning and Design Institute of Shaanxi Provincial Land Engineering Construction Group, Xi'an 710075, China

³ School of Traffic and Transportation, Shijiazhuang Tiedao University, Shijiazhuang 050043, China

managers to rapidly predict the spread of fires, providing a decision basis for formulating effective fire extinguishing plans. Forest fire behavior is a complex physical and chemical process. Extensive research has shown that the main factors affecting the spread of forest fires are forest flammability, weather, and terrain.

In the 1920s, Hawley (1926) and Gisborne (1927) suggested that fires could be simulated by observing the relationship between observed fire behavior and influencing factors. Fons (1946) proposed a physical forest fire model to measure and analyze fire behavior. Since then, many other models have been developed for investigating forest fire characteristics, including the Albini model (Albini 1979), elliptical model (Anderson et al. 1982), McArthur model (McArthur 1966), Canada forest fire model (Stocks et al. 1989), Rothermel model (Rothermel 1972), Van Wagner model (Van Wagner 1993), Wang model (Wang 1992), and so on (Sullivan 2009). Many computer simulation algorithms have also been developed, including the Huygens wave propagation method (Li et al. 2014), cellular automata (Duff et al. 2015), infiltration theory (Croft et al. 2016; Xia et al. 2009), fractal theory (Li et al. 2013), boundary interpolation, and the maze algorithm (Ding et al. 2014). These algorithms display forest fire spread using raster or vector forms.

These models have been used in practical applications through integration with various forest fire spread simulation tools or forest fire management systems, including FARSITE (Kanga and Singh 2017), Prometheus (D'ambrogio et al. 2016), and SiroFire (Papadopoulos and Pavlidou 2011). In these systems, a mathematical model quantifies forest fire behavior; then, the computer simulation algorithm combines the quantitative model with the characteristics of forest fire propagation to visualize predicted forest fire behavior. However, for large-scale forest fires in areas of complex terrain, meteorological conditions, or human factors, a significant error remains in the predicted temporal and spatial spread of fire (Finney et al. 2013).

The main causes of error include: (1) the accuracy of input data, in particular, that pertaining to the environment, terrain, and weather; and (2) a lack of clarity over the role of relevant parameters in the implementation of the algorithm (Wang et al. 2011); for example, the effect on both efficiency and accuracy of time steps in the cellular automata (CA) algorithm remains unclear.

In this study, the spatial distribution of vegetation in a forest fire area was classified using remote sensing data, while terrain information was extracted based on a digital elevation model (DEM). Based on actual fire data, the time correction of Wang (1992) was performed. Then, prediction of forest fire spread was realized through the cellular automata algorithm with variable step size in order to improve efficiency and accuracy. The results show that our modified algorithm can provide an efficient and accurate tool for emergency decision making.

2 Algorithm development

Our forest fire spreading algorithm consists of three parts: (1) extracting the distribution of plants, (2) the Wang (1992) time correction model, and (3) algorithm implementation by coupling CA with the improved model (Fig. 1).

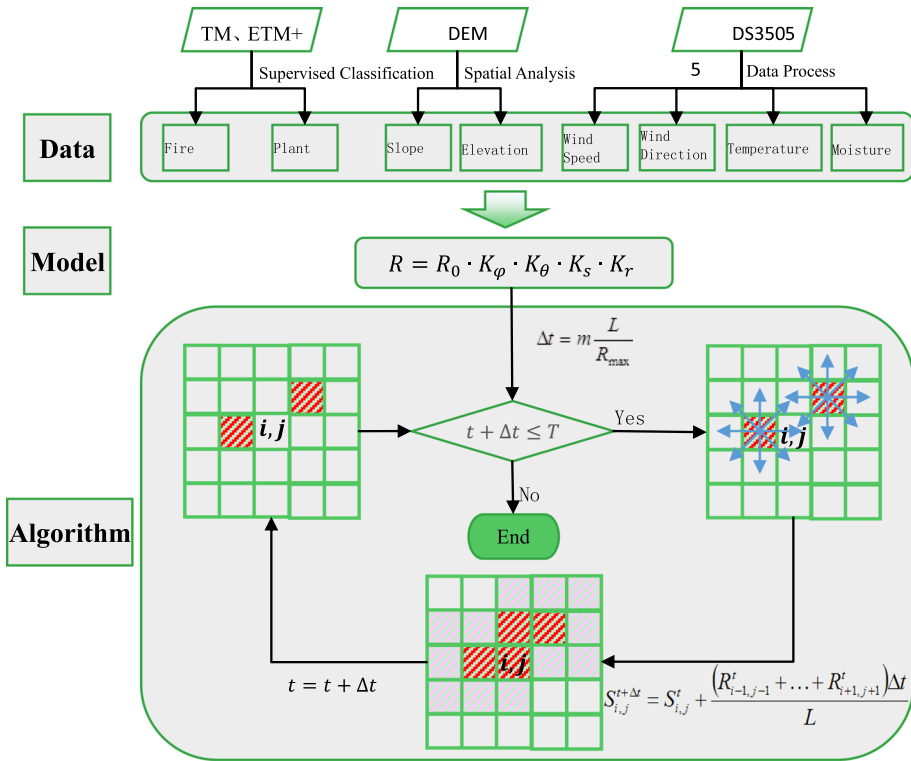


Fig. 1 Design of the forest fire spread algorithm based on geographic cellular automaton

2.1 Data description

Traditional forest fire propagation simulations have relatively more input parameters, which contributes to their lower efficiency (Li et al. 2012). In this study, weather station information and remote sensing images were selected as source data. Weather station data are DS3505 time-sharing data downloaded from the National Climatic Data Center (NCDC). ASTER GDEM version 2 and Landsat TM and ETM+ data were downloaded from the United States Geological Survey (USGS) Earth Explorer.

The study area was classified into pine trees, grass weeds, farmland, bare land, inhabited, rivers, and roads based on summertime remote sensing images analyzed using the maximum likelihood method. Combustible types were divided into pine, weeds, farmland, and others. According to Wang (1992), this two-step process is necessary to reduce errors caused by different spectral characteristics and to produce a more accurate surface vegetation classification map for simulating forest fire processes.

Fire data were based on remote sensing images, including fire location, smoke, and other features extracted using supervised classification according to different spectral and thermal characteristics (Zhan et al. 2012).

2.2 Time revised forest fire spread model

The forest fire spread model analyzes the influence of combustibles, wind, temperature, humidity, and slope on forest fire spread from physical, statistical, or empirical points. Based on the Wang (1992) and Mao (1993) models, the time correction coefficient was introduced to improve consistency between simulation and actual time, as expressed in Eq. (1):

$$R = R_0 \cdot K_\phi \cdot K_\theta \cdot K_s \cdot K_r \quad (1)$$

where

$$R_0 = a \cdot T + b \cdot W + c \cdot (100 - RH) - d,$$

$$W = \text{Int} \left(\frac{v}{0.836} \right)^{\frac{2}{3}},$$

$$K_\phi = e^{0.1783 \cdot v \cdot \cos \phi},$$

$$K_\theta = e^{3.553 \cdot g \cdot \tan(1.2 \cdot \theta)}$$

R is the speed of forest fire spread (m/min), R_0 is the initial speed of forest fire spread (m/min), K_ϕ is the wind coefficient, K_θ is a terrain factor, K_s is the combustible index (obtained from a lookup table; Wang 1992), K_r is the time correction coefficient (adjusted according to the actual fire), $a = 0.03$, $b = 0.05$, $c = 0.01$, $d = 0.3$, T is temperature ($^{\circ}\text{C}$), W is wind level, Int is a set integer, RH is air humidity (%), v is wind speed (m/s), ϕ is the angle between wind direction and fire spread direction, θ is slope, and g is the direction of the hill (1 for uphill and -1 for downhill).

The model takes into account the effects of combustibles, wind, temperature, humidity, and slope on the spread of forest fires. The time correction coefficient is based on spatial and temporal differences between simulated and actual fires, where a simulated fire occurs earlier than the actual fire, K_r should be reduced, and it should be increased when the actual fire occurs earlier.

2.3 Geographic cellular automata algorithm

Cellular automata are a dynamic system defined in cellular space by the cellular discrete finite-state composition, in accordance with local rules for certain discrete time evolution (Collin et al. 2011). It can be described as: $\langle n\text{-dimensional cellular space, the state, the number of neighborhoods, state transition rules} \rangle$ (i.e., $\langle Z^n, S, N, f \rangle$).

In the forest fire spread geographic CA algorithm, the geographic cell space (Z^n) consists of fire units with different geographic combustion conditions (S). According to the rules (f) of forest fire spread, the state of each cell changes with discrete time, and the state of each cell changes depending on its own state and the state of Moore neighborhood units (N), where the state transition rules $S^{t+\Delta t} = f(S^t, N)$ form the core of the entire system.

The resolution of most publicly downloadable remote sensing data is 30 m, and cells can be represented by grid pixels; therefore, our forest fire spread CA algorithm uses a 30-m grid image of the study area as the cell space, where the pixel value represents the cell state. Assuming S is the state of the cell, it can be described as follows: $S = 0$ (unburned), $S = 1$ (early burning), $S = 2$ (full burning, has the ability to ignite surrounding cells), $S = 3$ (extinguishing), or $S = 4$ (extinguished).

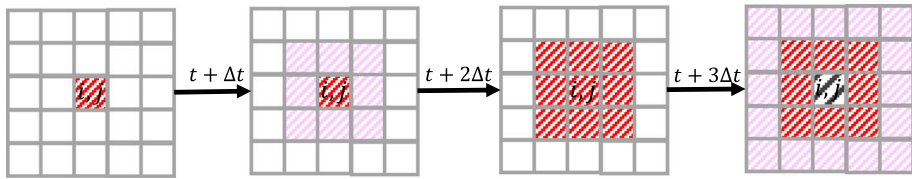


Fig. 2 Procedure for cellular state change

After inputting fire information and time, cells with a state of 2 will continuously ignite combustible neighborhood cells, controlling the effect of forest fire spread. The evolution of time is discrete; that is, after time step Δt , the state of the fire in the next moment $t + \Delta t$ can be calculated according to the fire state at time (t).

The state transition rule is designed based on actual forest fire procedures from early burning to extinguished. In the algorithm, if the state of a combustible cell is 0 ($S = 0$, unburned) and there exists $S = 1$ in neighborhood cells, then the cell value at the next time can be calculated by Eq. (2). If $S = 1$ at this time (early burning), then S will become 2 (full burning) at the next moment. If $S = 2$ and the states of neighborhood cells are equal or larger than 2 or are not combustible, then $S = 3$ (extinguishing gradually). If $S = 3$ at this time, then the state of the cell will be $S = 4$ at the next moment (completely extinguished). In the procedure of cellular state change (Fig. 2), cell (i, j) is the fire point that ignites surrounding neighborhood cells with a certain speed at time $t + \Delta t$. Neighborhood cells transition through early burning to full burning at time $t + 2\Delta t$, at which point cell (i, j) begins to extinguish. Neighborhood cells ignite their own neighborhood cells with a certain speed at time $t + 3\Delta t$. (In this algorithm, we assume the time taken to transition from one state to another is equal.) This process is expressed by the following equations:

$$S_{ij}^{t+\Delta t} = S_{ij}^t + \frac{(R_{i-1,j-1}^t + \dots + R_{i+1,j+1}^t)\Delta t}{L} \quad (R_{ij}^t = 0) \quad (2)$$

$$\Delta t = m \frac{L}{R_{\max}} \quad (m < 1) \quad (3)$$

where Δt is the time step; t is the current time; $t + \Delta t$ is the next moment; (i, j) are the column and row numbers of the cell, which represent cell location; $S_{ij}^{t+\Delta t}$ is the state of cell (i, j) at the next moment; $R_{i-1,j-1}^t$ is the speed of spread from the neighborhood cell $(i - 1, j - 1)$ to the central cell at time t ; $R_{\max}$ is the maximum spreading speed ($R_{\max}$ and Δt differ depending on the time and environmental conditions); m is the step size factor, where the higher the value, the higher the efficiency of the algorithm and the lower the simulation accuracy; and $L = 30$ represents the cell size.

3 Algorithm implementation

We developed the forest fire simulation system using ArcGIS Engine and C#. The simulation interval was 2 s, which is equal to an actual time of 30 min. The state of the fire is displayed after every 30 loops.

3.1 Data process

To develop our simulation model, we used a fire accident that occurred between May 23 and June 2, 2006, in the Daxing'anling area ($\sim 125.67\text{E}$ – 127.22E , $\sim 50.42\text{N}$ – 51.34N) of northeastern China. The ETM+ 5 7 band image of the study area is almost unaffected by smoke; however, the fire area is clearly visible (Fig. 3). Two images from this time period (May 27 at 10:30 using the Landsat5 TM and May 28 at 10:21 using the Landsat7 ETM+) were used to represent the start and end of fire spread. Landsat data from May 20 and 27, along with ASTER GDEM version 2 data from May 28, 2006, and DS3505 weather station data from between May 23 and June 2 were also obtained. After being transformed, interpreted, clipped, and resampled, the remote sensing images were used for vegetation classification with a resolution of 30 m and to identify the fire situation at two time points. Elevation and slope were obtained from DEM analysis. Meteorological data were converted into .xls format.

3.2 Optimal time step and time correction coefficient

To study the impact of time step on the simulation of forest fire spread, we set the forest fire spread speed to a constant. Taking a fire point as the initial fire source, the forest fire should spread into outward concentric circles. Step size m in the formula $\Delta t = m \frac{L}{R_{\max}}$ was set to 0.01, 0.03, 0.1, 0.125, 0.3, 0.5, 1.0, and 1.2. The results (Fig. 4) show that when m is 0.125, the analog graph is very close to a circle. However, when $m = 0.01, 0.03$, or 0.1 , the simulation result did not become more round even though the step size was very small. On that basis, we identified $m = 0.125$ as the optimal step size. When $m = 0.125 = 1.0/8.0$, the denominator is 8, which reflects the contribution of eight neighborhood cells to central cell combustion. The disadvantageous effects of smaller m on the results are greater than the benefits; for example, multiplied by $1.0/16.0$, accuracy is higher, but the number of cycles of all cell state values in the study area is multiplied and the algorithm rapidly loses

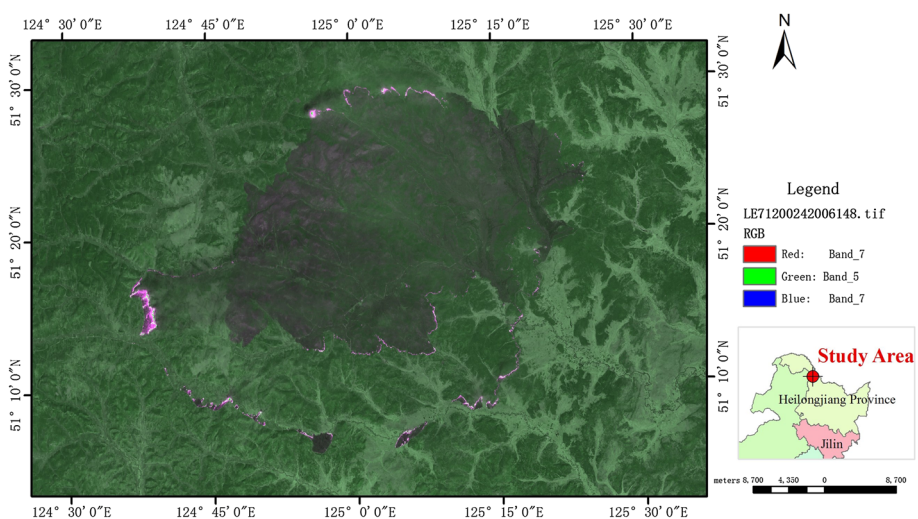


Fig. 3 Daxing'an Mountain forest fire area in 2006

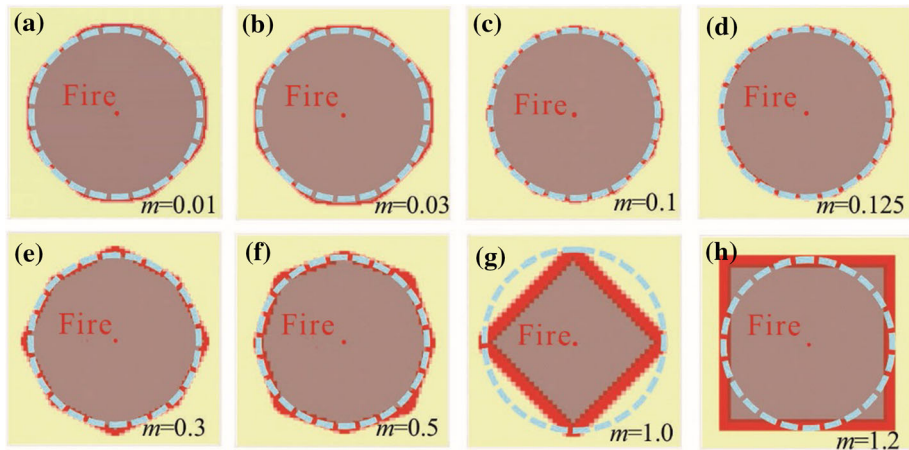


Fig. 4 Point fire spread simulation results for different time step coefficients (m): **a** $m = 0.01$; **b** $m = 0.03$; **c** $m = 0.1$; **d** $m = 0.125$; **e** $m = 0.3$; **f** $m = 0.5$; **g** $m = 1.0$; and **h** $m = 1$

efficiency. Our optimal time step represents $1/\text{neighborhood number}$, which is suitable for quadrilateral and hexagonal cells.

The results also show that without the time correction factor (i.e., $K_t = 1$), the forest fire spreads too quickly. For example, in fire zone 1 (Fig. 5), the simulated fire extent at 17:27

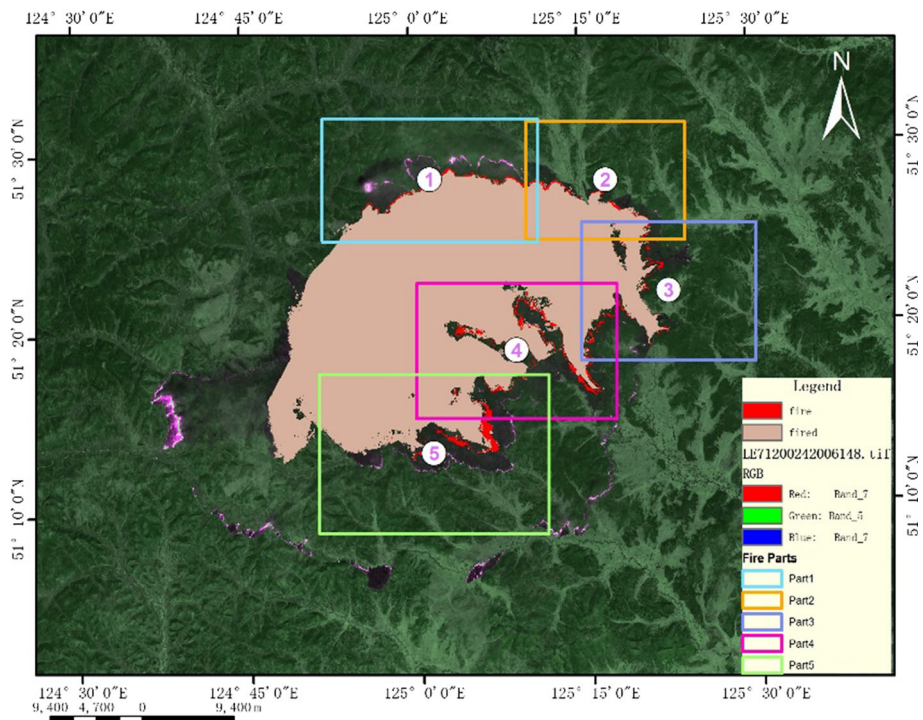


Fig. 5 Partition map for evaluating the accuracy of the forest fire spread simulation

Table 1 Time correction factor (K_r) values for fire partition zones

Zone	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Whole
K_r	0.3704	0.1667	0.1667	0.2000	0.3333	0.3448

See Fig. 5 for partition images

on May 27 equals the observed fire extent at 10:21 on May 28. When $K_r = 1/2.7$, the simulated fire spread speed was consistent with actual forest fire spread. However, we found that the optimal K_r value is location dependent (Table 1).

3.3 Simulation results and accuracy

In order to evaluate the accuracy of the algorithm, the fire area was divided into five fire zones (Fig. 5). By overlaying remote sensing images from 10:21 on May 28 over inferred fire start location on May 27 and the simulated fire at 10:21 on May 28 (Fig. 6), we found that simulation results for all five zones are almost coincident with the real fire in terms of shape and position. Accuracy evaluation (Table 2) shows that the Kappa coefficient is between 0.4003 and 0.7457 (mean = 0.6352), indicating high consistency, while the accuracy rate is between 84.56 and 93.20% (mean = 87.89%), indicating high accuracy. However, the simulation accuracy of the whole area was relatively low because of a new fire line in the south of the fire area. This fire line may represent a flying fire, a factor not considered by our algorithm. In addition, local wind, caused by the interaction between wind and rough terrain, is another factor not considered in the model. As zone 1 is located in rough terrain, and terrain can change wind direction, this may affect the accuracy. Finally, a reversible simulation algorithm of forest fire was realized to test whether the algorithm can accurately reverse the fire area to the fire point source after a passage of time. The results show that the reverse calculated location of the simulated fire point source is accurate (Fig. 7).

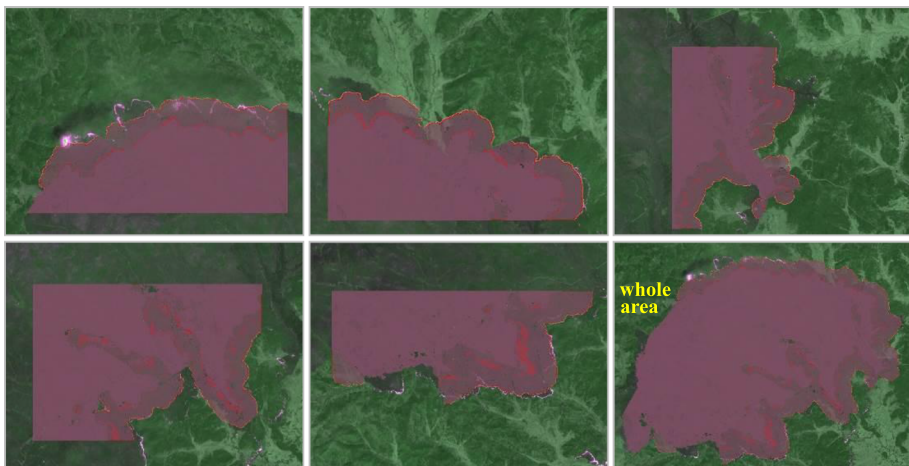
**Fig. 6** Overlay of simulated fire with real fire for each part of the study area

Table 2 Simulation fire situation accuracy

Area	Total pixels	Correct no. pixels	Simulated fire count	Simulated non-fire count	Actual fire count	Actual non-fire count	Accuracy (%)	Kappa
1	194,306	164,310	57,668	136,638	39,570	154,736	84.56	0.5933
2	160,964	150,014	27,194	133,770	23,960	137,004	93.20	0.7457
3	211,986	189,045	50,375	161,611	45,504	166,482	89.18	0.691
4	126,859	109,745	81,415	45,444	87,463	39,396	86.51	0.6977
5	301,304	262,502	90,442	210,862	80,188	221,116	87.12	0.6832
All	3,865,828	3,354,939	640,828	3,225,000	315,823	3,550,005	86.78	0.4003

For all parts, mean accuracy is 87.89% and Kappa coefficient is 0.6352. For parts 1–5, mean accuracy is 88.11% and Kappa coefficient is 0.6822

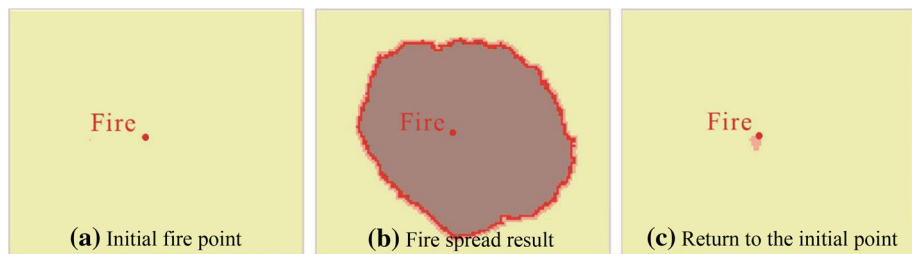


Fig. 7 Simulated fire spread **a** from the initial point, **b** to the maximum spread, and **c** back to the initial point

4 Conclusions

In this study, an improved forest fire spread simulation algorithm was developed. The algorithm was designed to ensure that data are easy to obtain, the model can be widely applied, and the parameters are adjustable. By coupling with the cellular automata algorithm, our approach improves on that of Wang (1992), which is one of the most popular models in China. The main findings of this study are as follows:

1. The simulation results show that forest fire spread is highly consistent with actual forest fire boundaries and the fire space. The Kappa coefficient is 0.6352 and mean accuracy is 87.89%, showing that both efficiency and accuracy are high. In addition, the algorithm is reversible, which means that it can find the original fire point using fire area data. This feature will be particularly useful for administrators working to identify the cause of a fire.
2. The spatial distribution of combustible materials, vegetation growth status, terrain, and other information sourced from remote sensing images are important factors that impact the accuracy of the algorithm. In the case of sparse meteorological data, the algorithm could be improved; in particular, optimization of the forest fire model

- correction and cell automaton time step could improve both the efficiency and accuracy of simulation results.
3. Through comparison of observed and simulated forest fire spread, simulation results were improved by optimizing the time step and time correction coefficient. The time consistency of simulated large-scale forest fire spread still requires improvement, for example, by incorporating meteorological data with higher temporal resolution and the assimilation of ground data to obtain more accurate meteorological data in time and space. Earth observation systems have improved rapidly in recent years (e.g., China's "high resolution satellite program"), and these will provide high temporal–spatial resolution images that will help in further studies of forest fire spread.
 4. No significant relationship was found between K_r and the normalized difference vegetation index (NDVI); however, the relationship between K_r and other factors should be taken into consideration in future research.

Acknowledgements This study was supported by the National Key Research and Development Program (Grant No: 2017YFB0503600) the National Natural Science Foundation of China (Grant No. 41771478), the Beijing Natural Science Foundation (Grant No. 8172046), and the Hebei Province Natural Science Fund (Grant No. D2016210008).

References

- Albini FA (1979) Spot fire distance from burning trees: a predictive model. Intermountain forest and range experiment station, forest service, US Department of Agriculture
- Anderson DH, Catchpole EA, De Mestre NJ et al (1982) Modelling the spread of grass fires. *J Aust Math Soc* 23(4):451–466
- Collin A, Bernardin D, Sero-Guillaume O (2011) A physical-based cellular automaton model for forest-fire propagation. *Combust Sci Technol* 183(4):347–369
- Croft P, Hunter JT, Reid N (2016) Forgotten fauna: habitat attributes of long-unburnt open forests and woodlands dictate a rethink of fire management theory and practice. *For Ecol Manage* 366:166–174
- Cunbin L, Jing Z, Baoguo T et al (2014) Analysis of forest fire spread trend surrounding transmission line based on rothermel model and huygens principle. *Int J Multimedia Ubiquitous Eng* 9(9):51–60
- D'ambrogio A, Gaudio P, Gelfusa M et al (2016) Use of integrated technologies for fire monitoring and first alert. In: *Proceedings of the 2016 IEEE 10th international conference on application of information and communication technologies (AICT)[C]*. IEEE Press, New York, pp 1–5
- Ding Z, Song K, Wang Z et al (2014) Contrastive analysis of algorithms of forest fire spreading simulation based on ArcGIS engine. *J Univ Chin Acad Sci* 31(5):640–646 (in Chinese)
- Duff TJ, Chong DM, Tolhurst KG (2015) Using discrete event simulation cellular automata models to determine multi-mode travel times and routes of terrestrial suppression resources to wildland fires. *Eur J Oper Res* 241(3):763–770
- Finney MA, Cohen JD, Mcallister SS et al (2013) On the need for a theory of wildland fire spread. *Int J Wildl Fire* 22(1):25–36
- Fons WL (1946) Analysis of fire spread in hght forest fuels. *J Agric Res* 72(3):93–121
- Gisborne HT (1927) The objectives of forest fire-weather research. *J Forest* 25(4):452–456
- Hawley LF (1926) Theoretical considerations regarding factors which influence forest fires. *J Forest* 24(7):756–763
- Kanga S, Singh SK (2017) Forest fire simulation modeling using remote sensing & GIS. *Int J Adv Res Comput Sci* 8(5):326–332
- Li D, Wang C, Hu Y et al (2012) General review on remote sensing-based biomass estimation. *Geom Inf Sci Wuhan Univ* 37(6):631–635 (in Chinese)
- Li F, Liu H, Zhang X et al (2013) Determination of spontaneous combustion extent in coal seams on the basis of the fractal theory. *Coal Geol Explor* 41(3):15–17 (in Chinese)
- Mao X (1993) The influence of wind and relief on the speed of the forest fire spreading. *Q J Appl Meteorol* 4(1):100–104 (in Chinese)

- McArthur AG (1966) Weather and grassland fire behaviour. Forestry and timber bureau, Department of National Development, Commonwealth of Australia
- Papadopoulos GD, Pavlidou F-N (2011) A comparative review on wildfire simulators. *Syst J IEEE* 5(2):233–243
- Rothermel RC (1972) A mathematical model for predicting fire spread in wildland fuels. USFS
- Stocks BJ, Lynham T, Lawson B et al (1989) Canadian forest fire danger rating system: an overview. *For Chron* 65(4):258–265
- Sullivan AL (2009) Wildland surface fire spread modelling, 1990–2007. 3: simulation and mathematical analogue models. *Int J Wildl Fire* 18(4):387–403
- Wagner CV (1993) Prediction of crown fire behavior in two stands of jack pine. *Can J For Res* 23(3):442–449
- Wang Z (1992) Current forest fire danger rating system. *J Nat Disasters* 1(3):39–44 **(in Chinese)**
- Wang H, Zhang W, Chen Y et al (2011) Fire spreading model based on CA scope. *Geom Inf Sci Wuhan Univ* 36(5):575–578 **(in Chinese)**
- Xia Q, Li G, Yin J (2009) Application of percolation theory in forest fire simulation. *J Zhejiang For Coll* 26(2):233–238 **(in Chinese)**
- Zhan J, Fan C, Li T (2012) A forest firespot automatic detection algorithm for HJ-IRS imagery. *Geom Inf Sci Wuhan Univ* 37(11):1321–1324 **(in Chinese)**